

# Supplementary Material for CoFiTok: Coarse-to-Fine Denoising Tokens for Pixel-Space Diffusion

Anonymous submission

## Supplementary Scope

This document separates comparison evidence from the main paper’s mechanism-centered tables by protocol, not by outcome. P0 contains matched-dataset and matched-step pixel-space training. P1 contains official-pretrained tokenizer reconstruction. A third tier contains official ImageNet-256 50K evaluations for architecture-incompatible methods. Values from different tiers are not used to claim direct superiority.

## Complete Evidence Coverage

Table S2 exposes the full eight-dataset by 15-method Cartesian product. A blocked cell marks a cross-task comparison that is not interchangeable with the available official evaluation; official results are never counted as matched training.

The locked matrix contains 80 completed P0 train/eval cells, 16 P1 eval-only cells, and 24 protocol-blocked cells. Three official 50K rows remain separate.

## External-Method Results

The next four tables are generated from the locked evidence JSON. They report P0 generation, FlexTok/TiTok reconstruction, official D-AR/MAR/ReTok ImageNet-256 results, and the paper’s claim boundary.

## How to Read the External Comparisons

**Improved DDPM and EDM.** Table S3 is the direct external pixel-diffusion context. Dataset, resolution, split, and nominal optimizer steps are aligned, but parameter count, batch size, nominal images seen, and NFE remain explicit. EDM is the low-resolution Frechet-style winner on seven of eight rows, and the direct dense predictor wins the remaining row. These results rule out a broad unconditional-generation superiority claim for CoFiTok. They do not test the paper’s distinctive interface claim because monolithic predictors do not expose ordered component prefixes.

**FlexTok and TiTok.** Table S4 uses official-pretrained models and evaluates reconstruction at  $256 \times 256$  on 256 images per dataset. These methods tokenize image content; CoFiTok factorizes a dense diffusion noise prediction. Their reconstruction PSNR and Frechet-style values therefore document adjacent capability rather than a shared generation objective.

**D-AR, MAR, and ReTok.** Table S5 reports 50K-sample official evaluations on ImageNet-256. The rows use available pretrained checkpoints and their corresponding generation protocols. They are useful large-model context, but they cannot be relabeled as eight-dataset matched-step retraining. The Inception Score for MAR follows the ADM-style evaluator used in the recorded run and is not mixed with the differently scaled value reported by another evaluation implementation.

## Matched P0 Protocol Details

The P0 protocol fixes dataset identity, spatial resolution, split, and nominal optimizer steps. It does not pretend that different published architectures have equal FLOPs. Table S7 exposes the remaining major differences instead of hiding them in a single rank.

Only the direct dense control isolates factorization at nearly equal parameter count. Improved DDPM and EDM test whether the short-budget generator is competitive with established monolithic objectives, while endpoint-only factorized prediction tests whether ordered prefix supervision contributes beyond retaining  $K$  token components.

---

This is an anonymized submission for review purposes only. Distribution, citation, or public sharing of this manuscript is strictly prohibited. Copyright and publication details will appear in the final version if accepted.

Table S1: Comparison protocols used throughout the paper and supplement.

| Tier                | Methods                                                             | Coverage       | Evidence                                         | Valid interpretation                                                      |
|---------------------|---------------------------------------------------------------------|----------------|--------------------------------------------------|---------------------------------------------------------------------------|
| P0 matched training | CoFiTok, direct dense, endpoint-only, ablations, Improved DDPM, EDM | Eight datasets | Train/eval under locked per-dataset steps        | Within-row short-budget context; external methods are not compute matched |
| P1 reconstruction   | FlexTok, TiTok                                                      | Eight datasets | Official-pretrained reconstruction on 256 images | Tokenizer reconstruction context only; not a generation comparison        |
| Official related    | D-AR, MAR, ReTok                                                    | ImageNet-256   | Official-pretrained 50K evaluation               | Secondary large-model context; not matched retraining against CoFiTok     |

Table S2: Evidence coverage matrix (8 datasets  $\times$  15 methods). T = completed matched train/eval; E = completed official-pretrained reconstruction eval-only; B = matched cross-task protocol blocked; O = separate official ImageNet-256 50K eval-only result. B+O retains the blocked matched cell while exposing the secondary official result.

| Method                         | CIFAR | Tiny-IN | IN64-HF | IN64-strict | FFHQ | AFHQv2 | IN256-10% | IN256 |
|--------------------------------|-------|---------|---------|-------------|------|--------|-----------|-------|
| CoFiTok                        | T     | T       | T       | T           | T    | T      | T         | T     |
| Direct dense epsilon           | T     | T       | T       | T           | T    | T      | T         | T     |
| Endpoint-only factorized       | T     | T       | T       | T           | T    | T      | T         | T     |
| Legacy clean-prefix objective  | T     | T       | T       | T           | T    | T      | T         | T     |
| No denoise-path prefix term    | T     | T       | T       | T           | T    | T      | T         | T     |
| Add clean-space monotonic term | T     | T       | T       | T           | T    | T      | T         | T     |
| Simultaneous prediction        | T     | T       | T       | T           | T    | T      | T         | T     |
| Deep- $S_k$ synthesis          | T     | T       | T       | T           | T    | T      | T         | T     |
| Improved DDPM                  | T     | T       | T       | T           | T    | T      | T         | T     |
| EDM                            | T     | T       | T       | T           | T    | T      | T         | T     |
| D-AR                           | B     | B       | B       | B           | B    | B      | B         | B+O   |
| FlexTok                        | E     | E       | E       | E           | E    | E      | E         | E     |
| TiTok                          | E     | E       | E       | E           | E    | E      | E         | E     |
| MAR                            | B     | B       | B       | B           | B    | B      | B         | B+O   |
| ReTok                          | B     | B       | B       | B           | B    | B      | B         | B+O   |

## Dataset and Source Conditions

The two ImageNet-64 aliases are intentionally distinct. Results labeled ImageNet-64 HF use the HF-derived source; results labeled ImageNet-64 strict use the exact downsampled ImageNet source. No result is silently renamed between them.

## Metric Definitions

For prefix budget  $m$ , the cumulative prediction and pixel-space estimate are

$$\hat{\epsilon}^{(1:m)} = \sum_{k=1}^m S_k(z^{(k)}), \quad \hat{x}_0^{(m)} = \frac{x_t - \sigma_t \hat{\epsilon}^{(1:m)}}{\alpha_t}. \quad (1)$$

Let  $e_m$  be the mean-squared error between  $\hat{x}_0^{(m)}$  and the preregistered denoise-path target at prefix  $m$ . The reported path AUC is the equal-spaced trapezoid average

$$\text{AUC}_{\text{path}} = \frac{1}{K-1} \sum_{m=1}^{K-1} \frac{e_m + e_{m+1}}{2}. \quad (2)$$

Lower values indicate that intermediate prefixes follow the intended partial denoising path. Endpoint MSE instead uses only  $m = K$  and does not measure the quality of intermediate prefixes.

For component energy proportions  $p_k$ , effective token count is

$$K_{\text{eff}} = \exp\left(-\sum_k p_k \log p_k\right). \quad (3)$$

The zero-token ratio divides the energy produced by  $S_k(0)$  by the energy of the corresponding observed components. A bias-free restricted synthesis operator satisfies this diagnostic exactly up to numerical precision.

## Repeated and Confirmatory Evidence

The matched 20K repeat uses K8 models, seeds 103 and 139, and 1,024 validation images at  $t = 500$  on Tiny ImageNet-200 and ImageNet-64 HF. CoFiTok lowers path AUC in all four paired dataset-seed comparisons, with a mean relative reduction of 96.18%. The mean endpoint  $x_0$ -MSE change is +4.20%, so this repeat supports prefix organization rather than an endpoint-quality win.

The ImageNet-256 K4 20K confirmatory run uses the same two seeds and exhaustively evaluates all 24 component permutations. The learned order ranks first for both seeds, the paired-bootstrap lower bounds for non-identity minus ordered AUC are positive, and every permutation preserves the endpoint component sum up to  $3 \times 10^{-15}$  MSE. Relative to the parameter-matched direct dense control, CoFiTok lowers endpoint MSE by 6.73% on average in this confirmatory setting.

## Matched Component Ablations

The component ablation suite uses ImageNet-64 HF, K8, 5,000 optimizer steps, seeds 103/139, and 1,024 validation images at fixed  $t = 500$ . All variants share the data, model width, optimizer, and evaluation noise for each seed. The

Table S3: Eight-dataset P0 generated-sample comparison under the locked short-budget protocols. Lower Frechet-style metrics are better.

| Dataset            | C low | D low | DDPM low | EDM low | low winner           | C Inc  | D Inc  | DDPM Inc | EDM Inc | Inc winner           | C rank |
|--------------------|-------|-------|----------|---------|----------------------|--------|--------|----------|---------|----------------------|--------|
| CIFAR-10           | 6.47  | 5.80  | 7.86     | 2.87    | EDM                  | 312.00 | 359.50 | 498.65   | 351.17  | CoFiTok              | 1      |
| Tiny-IN            | 8.49  | 6.99  | 9.07     | 3.10    | EDM                  | 373.63 | 392.40 | 381.24   | 268.72  | EDM                  | 2      |
| ImageNet-64 HF     | 8.09  | 6.89  | 8.88     | 3.23    | EDM                  | 390.99 | 417.79 | 378.33   | 258.40  | EDM                  | 3      |
| ImageNet-64 strict | 8.36  | 6.87  | 8.80     | 2.75    | EDM                  | 374.68 | 383.72 | 355.11   | 248.31  | EDM                  | 3      |
| FFHQ-64            | 8.74  | 6.84  | 9.44     | 2.51    | EDM                  | 425.71 | 427.05 | 369.01   | 325.49  | EDM                  | 3      |
| AFHQv2-64          | 7.08  | 5.68  | 8.29     | 1.96    | EDM                  | 331.06 | 333.42 | 335.18   | 230.40  | EDM                  | 2      |
| ImageNet-256 10%   | 10.44 | 12.59 | 10.40    | 10.39   | EDM                  | 414.19 | 366.42 | 419.60   | 368.68  | Direct dense epsilon | 3      |
| ImageNet-256       | 10.54 | 10.24 | 10.40    | 10.40   | Direct dense epsilon | 417.26 | 354.13 | 414.30   | 368.43  | Direct dense epsilon | 4      |

C = CoFiTok; D = parameter-matched direct dense epsilon. Rows use the locked per-dataset short-budget protocol; they are not formal SOTA FID claims.

Table S4: Official-pretrained tokenizer reconstruction context (eval-only).

| Dataset            | src | eval | FlexTok PSNR | FlexTok low | TiTok PSNR | TiTok low |
|--------------------|-----|------|--------------|-------------|------------|-----------|
| AFHQv2-64          | 64  | 256  | 24.04        | 0.036       | 17.29      | 0.217     |
| CIFAR-10           | 32  | 256  | 26.77        | 0.037       | 19.35      | 0.168     |
| ImageNet-64 strict | 64  | 256  | 22.57        | 0.042       | 16.73      | 0.196     |
| FFHQ-64            | 64  | 256  | 24.06        | 0.042       | 17.39      | 0.231     |
| ImageNet-64 HF     | 64  | 256  | 23.60        | 0.039       | 18.04      | 0.182     |
| ImageNet-256       | 256 | 256  | 19.12        | 0.044       | 15.54      | 0.196     |
| ImageNet-256 10%   | 256 | 256  | 19.12        | 0.044       | 15.54      | 0.196     |
| Tiny-IN            | 64  | 256  | 21.81        | 0.044       | 17.30      | 0.160     |

Official pretrained reconstruction at 256x256 on 256 images per dataset. These rows are not retrained generation baselines.

progress-power sweep is scored against one common  $p = 1.5$  path; it is not scored against a different target for each point.

The direct removals distinguish the path-prefix and path-component terms from the endpoint objective. The simultaneous variant removes token-to-token feedback while retaining the same heads and losses. The deep-synthesis variant retains token-only inputs but introduces biased nonlinear layers, so its zero-token ratio directly tests whether decoder capacity can carry a token-independent prior. Full CoFiTok lowers path AUC by 93.0% relative to endpoint-only and by 45.7%/37.9% relative to removing the prefix/component path term, respectively. Endpoint MSE changes by less than 0.2% in the two single-term comparisons. Sequential feedback changes the mean more modestly ( $-2.8\%$  path AUC and  $-1.0\%$  endpoint MSE) and wins path AUC on only one of two seeds, so it is treated as an optional predictor implementation rather than part of the core claim. Deep synthesis reaches lower path AUC but produces a nonzero zero-token ratio of  $0.0485 \pm 0.0050$ . The parameter-matched direct dense control is only 0.6% lower in endpoint MSE at 64px but exposes no multi-prefix interface; conversely, the ImageNet-256 con-

firmary setting gives CoFiTok a 6.73% endpoint advantage over direct dense. Factorization is therefore supported as an interface with a small or favorable endpoint tradeoff, not as a universal endpoint-quality guarantee.

### Hyperparameter Sensitivity

Figure S1 and Table S12 report the full two-seed sweeps. The dashed lines mark the main values:  $\lambda_p = 0.15$ ,  $\lambda_c = 0.30$ ,  $p = 1.5$ , and  $K = 8$ . Increasing either loss weight improves path AUC with little endpoint movement; the chosen weights are a moderate endpoint/path tradeoff. Power 1.5 is the minimum under the common evaluation target. K8 has the lowest mean path AUC, K4 the lowest endpoint MSE, and K16 no short-budget path advantage.

### Evidence Lock and Reproducibility

All numerical tables imported above are generated from the final locked JSON evidence package. The paper source does not hand-edit those metrics. The locked state contains the per-run configuration, dataset alias, split, optimizer steps, batch size, seed, parameter count, NFE, evaluation image count,

Table S5: Official ImageNet-256 related-method evaluations (50K samples).

| Method | Dataset      | Protocol               | Samples | FID   | sFID  | IS     | Prec. | Rec.  |
|--------|--------------|------------------------|---------|-------|-------|--------|-------|-------|
| D-AR   | ImageNet-256 | official 50K eval-only | 50000   | 2.628 | 6.720 | 285.89 | 0.803 | 0.588 |
| MAR    | ImageNet-256 | official 50K eval-only | 50000   | 2.338 | 4.700 | 62.90  | 0.822 | 0.572 |
| ReTok  | ImageNet-256 | official 50K eval-only | 50000   | 2.219 | 5.896 | 245.94 | 0.816 | 0.599 |

These rows use official pretrained protocols where available and are secondary comparisons; they are not part of the matched-dataset/optimizer-step P0 matrix.

Table S6: Claim-evidence boundary.

| Claim axis            | Evidence summary                                                                                                                                                                      |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Generation quality    | CoFiTok is best on 0/8 lowres rows and 1/8 Inception rows; average ranks are 3.12 and 2.62. Do not claim generation SOTA.                                                             |
| Prefix control        | CoFiTok path AUC is lower than the endpoint-only factorized control on 8/8 datasets.                                                                                                  |
| Repeated 20k evidence | CoFiTok lowers path AUC in 4/4 paired dataset-seed runs (mean relative reduction 96.18%), while endpoint $x_0$ MSE at $t = 500$ changes by +4.20% on average.                         |
| ImageNet-256 scaling  | ImageNet-256 K4 20k confirmatory gates: 7/7 passed; overall PASS; mean endpoint-MSE change -6.73%; ordered ranks among all 24 permutations 1/1.                                       |
| Restricted synthesis  | CoFiTok zero-token ratio is zero in current rows; deep- $S_k$ has nonzero zero-token ratio on 8/8 datasets.                                                                           |
| Related methods       | completed official 50K: D-AR, MAR, ReTok. All rows remain secondary and outside P0 matched-dataset/step training.                                                                     |
| Main paper stance     | The core empirical gates support the scoped claim of ordered restricted dense-noise factorization with prefix-controllable denoising, not broad unconditional generation superiority. |

and sample count needed to distinguish the three protocol tiers. New experiments must produce a new evidence package rather than silently overwrite these rows.

Table S7: Locked P0 short-budget settings. C = CoFiTok; D = direct dense; E = endpoint-only factorized. NFE is listed in the order C/D/Improved-DDPM/EDM.

| Protocol family | Datasets                                    | Resolution | Steps | C/D/E batch | DDPM/EDM batch | Quality images | Sample/real images | NFE         |
|-----------------|---------------------------------------------|------------|-------|-------------|----------------|----------------|--------------------|-------------|
| CIFAR-10        | CIFAR-10                                    | 32         | 3000  | 64          | 32             | 512            | 1024/4096          | 50/50/50/79 |
| 64px            | Tiny-IN, IN64-HF, IN64-strict, FFHQ, AFHQv2 | 64         | 5000  | 32          | 32             | 512            | 1024/4096          | 50/50/50/79 |
| ImageNet-256    | 10% and full train                          | 256        | 5000  | 4           | 4              | 256            | 512/2048           | 50/50/50/79 |

Table S8: Trainable parameter counts in the P0 protocols. The direct dense predictor is the parameter-matched factorization control; Improved DDPM and EDM are external architecture baselines.

| Resolution       | CoFiTok | Direct dense | Dense delta vs. CoFiTok | Improved DDPM | EDM       |
|------------------|---------|--------------|-------------------------|---------------|-----------|
| $32 \times 32$   | 81,808  | 80,707       | −1.35%                  | 9,593,539     | 3,870,531 |
| $64 \times 64$   | 81,808  | 80,707       | −1.35%                  | 21,715,907    | 3,870,531 |
| $256 \times 256$ | 15,776  | 15,690       | −0.55%                  | 7,200,579     | 952,291   |

Table S9: Dataset roles in the locked evidence matrix.

| Dataset label      | Resolution | Split role                | Source condition                                            |
|--------------------|------------|---------------------------|-------------------------------------------------------------|
| CIFAR-10           | 32         | Train/test                | Canonical CIFAR-10 staging                                  |
| Tiny-IN            | 64         | Train/validation          | Tiny ImageNet-200 staging                                   |
| ImageNet-64 HF     | 64         | Train/validation          | HF-derived ImageNet-1K 64x64 source                         |
| ImageNet-64 strict | 64         | Train/validation          | Exact downsampled ImageNet source from Academic Torrents    |
| FFHQ-64            | 64         | Train/evaluation          | Non-classification face-domain extension                    |
| AFHQv2-64          | 64         | Train/evaluation          | Non-classification animal-domain extension                  |
| ImageNet-256 10%   | 256        | 10% train/full validation | Deterministic subset derived from the staged 256x256 source |
| ImageNet-256       | 256        | Full train/validation     | Full staged 256x256 ImageNet source                         |

Table S10: Matched component ablations (mean  $\pm$  sample standard deviation). Lower is better for endpoint MSE, path AUC, and zero ratio; higher effective K indicates broader component use.

| Variant                  | Endpoint $x_0$ MSE $\downarrow$ | Path AUC $\downarrow$ | Eff. K $\uparrow$ | Zero ratio $\downarrow$ |
|--------------------------|---------------------------------|-----------------------|-------------------|-------------------------|
| CoFiTok (full)           | $0.1300 \pm 0.0027$             | $0.0686 \pm 0.0017$   | $6.780 \pm 0.015$ | $0.0000 \pm 0.0000$     |
| Endpoint-only factorized | $0.1275 \pm 0.0019$             | $0.9991 \pm 0.1607$   | $5.366 \pm 0.054$ | $0.0000 \pm 0.0000$     |
| No path-prefix term      | $0.1300 \pm 0.0003$             | $0.1269 \pm 0.0096$   | $6.547 \pm 0.027$ | $0.0000 \pm 0.0000$     |
| No path-component term   | $0.1302 \pm 0.0012$             | $0.1105 \pm 0.0013$   | $6.752 \pm 0.069$ | $0.0000 \pm 0.0000$     |
| Simultaneous tokens      | $0.1314 \pm 0.0016$             | $0.0709 \pm 0.0081$   | $6.829 \pm 0.025$ | $0.0000 \pm 0.0000$     |
| Deep synthesis           | $0.1496 \pm 0.0305$             | $0.0463 \pm 0.0051$   | $6.774 \pm 0.032$ | $0.0485 \pm 0.0050$     |

Table S11: Parameter-matched architecture control under the same two-seed 5k ImageNet-64 HF protocol. Direct dense has no native multi-prefix path.

| Variant                        | Endpoint $x_0$ MSE $\downarrow$ | Path AUC $\downarrow$ | Eff. K $\uparrow$ | Zero ratio $\downarrow$ |
|--------------------------------|---------------------------------|-----------------------|-------------------|-------------------------|
| CoFiTok (full)                 | $0.1300 \pm 0.0027$             | $0.0686 \pm 0.0017$   | $6.780 \pm 0.015$ | $0.0000 \pm 0.0000$     |
| Parameter-matched direct dense | $0.1292 \pm 0.0006$             | —                     | $1.000 \pm 0.000$ | $0.0000 \pm 0.0000$     |

Matched ImageNet-64 ablations (mean and sample s.d.; seeds 103/139)

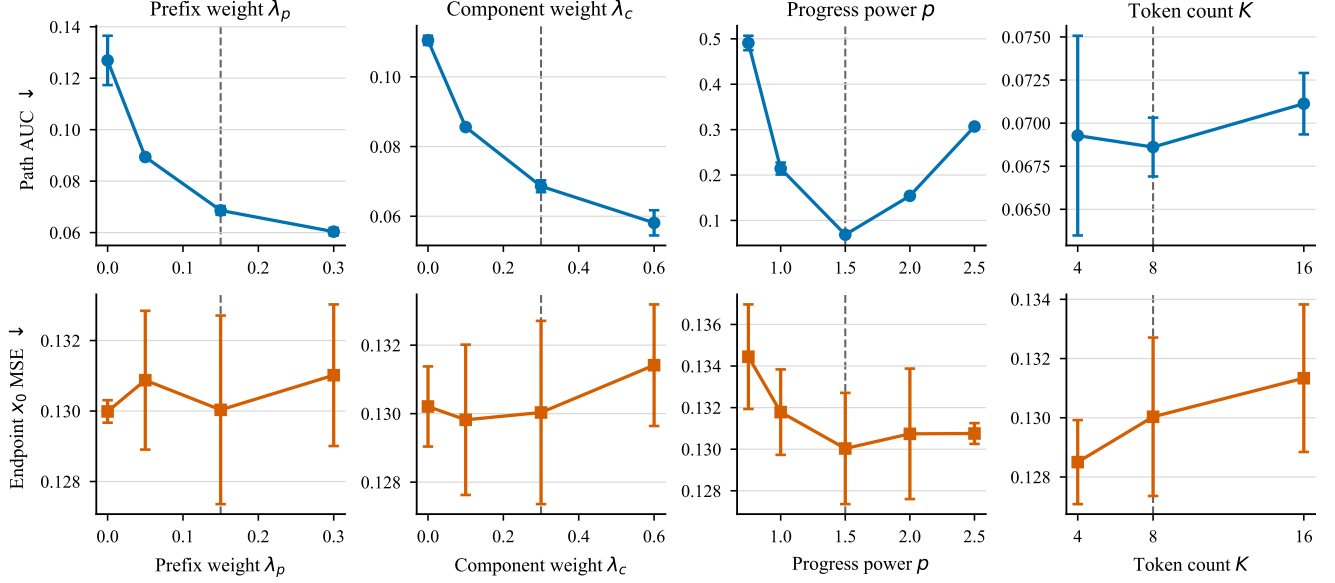


Figure S1: Matched ImageNet-64 HF hyperparameter sensitivity. Top row: denoise-path AUC; bottom row: endpoint  $x_0$  MSE. Error bars are sample standard deviations over seeds 103/139. Lower is better.

Table S12: Numerical values underlying Figure S1. Each entry is mean  $\pm$  sample standard deviation over seeds 103/139.

| Axis        | Value | Endpoint $x_0$ MSE $\downarrow$ | Path AUC $\downarrow$ | Eff. K $\uparrow$  |
|-------------|-------|---------------------------------|-----------------------|--------------------|
| $\lambda_p$ | 0     | 0.1300 $\pm$ 0.0003             | 0.1269 $\pm$ 0.0096   | 6.547 $\pm$ 0.027  |
|             | 0.05  | 0.1309 $\pm$ 0.0020             | 0.0894 $\pm$ 0.0009   | 6.694 $\pm$ 0.033  |
|             | 0.15  | 0.1300 $\pm$ 0.0027             | 0.0686 $\pm$ 0.0017   | 6.780 $\pm$ 0.015  |
|             | 0.3   | 0.1310 $\pm$ 0.0020             | 0.0604 $\pm$ 0.0015   | 6.819 $\pm$ 0.002  |
| $\lambda_c$ | 0     | 0.1302 $\pm$ 0.0012             | 0.1105 $\pm$ 0.0013   | 6.752 $\pm$ 0.069  |
|             | 0.1   | 0.1298 $\pm$ 0.0022             | 0.0856 $\pm$ 0.0004   | 6.860 $\pm$ 0.001  |
|             | 0.3   | 0.1300 $\pm$ 0.0027             | 0.0686 $\pm$ 0.0017   | 6.780 $\pm$ 0.015  |
|             | 0.6   | 0.1314 $\pm$ 0.0018             | 0.0581 $\pm$ 0.0036   | 6.717 $\pm$ 0.010  |
| $p$         | 0.75  | 0.1344 $\pm$ 0.0025             | 0.4909 $\pm$ 0.0159   | 7.035 $\pm$ 0.102  |
|             | 1     | 0.1318 $\pm$ 0.0021             | 0.2143 $\pm$ 0.0133   | 7.912 $\pm$ 0.015  |
|             | 1.5   | 0.1300 $\pm$ 0.0027             | 0.0686 $\pm$ 0.0017   | 6.780 $\pm$ 0.015  |
|             | 2     | 0.1307 $\pm$ 0.0031             | 0.1541 $\pm$ 0.0010   | 5.449 $\pm$ 0.020  |
|             | 2.5   | 0.1308 $\pm$ 0.0005             | 0.3069 $\pm$ 0.0031   | 4.459 $\pm$ 0.007  |
| $K$         | 4     | 0.1285 $\pm$ 0.0014             | 0.0693 $\pm$ 0.0058   | 3.321 $\pm$ 0.006  |
|             | 8     | 0.1300 $\pm$ 0.0027             | 0.0686 $\pm$ 0.0017   | 6.780 $\pm$ 0.015  |
|             | 16    | 0.1313 $\pm$ 0.0025             | 0.0711 $\pm$ 0.0018   | 13.676 $\pm$ 0.154 |

Table S13: Exploratory ImageNet-256 confirmatory generation controls. These rows do not replace the path and endpoint preregistered tests. Lower is better.

| Seed | Method                   | Samples | Lowres Frechet | Inception Frechet |
|------|--------------------------|---------|----------------|-------------------|
| 103  | Endpoint-only factorized | 4096    | 11.1619        | 370.3672          |
| 103  | Direct dense             | 4096    | 9.7396         | 449.3913          |
| 103  | CoFiTok                  | 4096    | 10.8232        | 349.4076          |
| 139  | Endpoint-only factorized | 4096    | 8.9753         | 404.6116          |
| 139  | Direct dense             | 4096    | 10.5597        | 384.8126          |
| 139  | CoFiTok                  | 4096    | 11.0839        | 356.9311          |